

# Empirical Predictions of Topological Spacetime Cosmology: Void Expansion Dynamics and Metric Anisotropy

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## Abstract

We present three falsifiable empirical predictions of Topological Spacetime Cosmology, a parameter-free geometric framework. The model derives cosmic acceleration strictly as a kinematic relaxation of the spatial metric as the macroscopic baryonic load dilutes, entirely eliminating the theoretical necessity for exogenous dark energy or dark matter parameters. We predict: (I) Void Expansion Dynamics: Deep within macroscopic cosmic voids, the local Hubble rate will bypass the standard average and converge explicitly to the unhindered topological metric baseline of  $H_{base} = 81.17$  km/s/Mpc. (II) Metric Anisotropy: The macroscopic expansion rate will exhibit severe anisotropy, acting as an exact inverse function of line-of-sight baryonic density, in stark contrast to the isotropic assumptions of  $\Lambda$ CDM. (III) Standard Sirens Baseline: Expansion rates derived strictly from pure metric propagation via gravitational waves (standard sirens) will bypass photonic-baryonic friction, trending significantly higher than photon-based measurements and converging toward the 81.17 metric baseline. These predictions are testable within the near term by upcoming missions and surveys such as Euclid, JWST, and LISA.

# 1. Introduction

Rather than attributing the accelerated expansion of the universe to a mysterious repulsive vacuum energy ( $\Lambda$ ), Topological Spacetime Cosmology dictates that late-time acceleration is a natural kinematic relaxation process. In this parameter-free framework, the pristine spatial metric inherently expands at a constant baseline rate of  $H_{base} = 81.17 \text{ km/s/Mpc}$ . The globally observed suppression of this rate (e.g.,  $H_0 \approx 73$ ) is exclusively a thermodynamic consequence of ordinary baryonic mass exerting a geometric load on the manifold.

Furthermore, the universal Time-Scaling Ratio ( $S_r \approx 7.03 \times 10^{60}$ ), extracted directly from this macroscopic baseline, acts as the absolute holographic scaling parameter that bounds both the vacuum energy density and the total baryonic mass of the observable universe ( $1.53 \times 10^{53} \text{ kg}$ ). Because this framework treats spacetime as a holographically bounded thermodynamic manifold rather than an infinitely divisible continuum, it fundamentally rejects the isotropic and homogeneous expansion rates modeled by  $\Lambda$ CDM. We present the following near-term predictive tests to empirically distinguish and validate this topological geometric framework.

## 2. Prediction I: Void Expansion Dynamics and Kinematic Unloading

**Current  $\Lambda$ CDM Paradigm:** The standard cosmological model assumes that late-time acceleration is driven by a uniform cosmological constant ( $\Lambda$ ) or vacuum energy. Consequently, the FLRW metric dictates that the expansion rate deep within cosmic voids should largely reflect the global average ( $H_0 \approx 67\text{--}73 \text{ km/s/Mpc}$ ), with minor deviations strictly attributed to local peculiar velocities.

**Topological Framework Prediction:** In Topological Spacetime Cosmology, late-time acceleration is not driven by an exogenous repulsive force, but rather emerges

as a natural kinematic relaxation of the spatial manifold as the macroscopic baryonic load dilutes. Therefore, where the baryonic stress-energy (topological resistance) is effectively zero, the spatial metric must relax to its pristine, unhindered expansion rate.

**Empirical Falsification Test:** Deep-space spectroscopic surveys (e.g., DESI, Euclid) measuring the local Hubble parameter deep within macroscopic underdense regions (cosmic voids) will detect a sharp, localized deviation from the global consensus. The expansion rate inside these voids will explicitly bypass the  $\Lambda$ CDM average and asymptotically converge to the unhindered topological baseline of  $H_{base} = 81.17$  km/s/Mpc. Confirming this localized kinematic acceleration in the absence of baryonic mass definitively falsifies the uniformity of dark energy.

### **3. Prediction II: Macroscopic Metric Anisotropy via Baryonic Load Inversion**

**Current  $\Lambda$ CDM Paradigm:** The cosmological principle necessitates that the universe is homogeneous and isotropic on large scales. The expansion rate is therefore expected to exhibit an isotropic profile, independent of the directional line-of-sight macroscopic density.

**Topological Framework Prediction:** Because the macroscopic expansion rate is fundamentally an inverse thermodynamic function of the local baryonic stress-energy deforming the manifold, the expansion of the universe cannot be isotropic. The metric dilation is severely restrained by the presence of baryonic mass.

**Empirical Falsification Test:** Comprehensive directional mapping of the Hubble parameter will reveal extreme macroscopic metric anisotropy that perfectly mirrors the structural density of the cosmic web. Sightlines intersecting massive baryonic concentrations, such as galactic superclusters, will yield heavily

suppressed expansion rates ( $H_0 \approx 67\text{-}70$  km/s/Mpc). Conversely, diametric sightlines traversing predominantly empty space will yield accelerated rates converging toward  $H_{base}$  (75-81 km/s/Mpc). Demonstrating that the expansion rate acts as an exact inverse function of the line-of-sight baryonic density proves that ordinary mass is the sole geometric load governing cosmic kinematics.

## 4. Prediction III: The Standard Sirens Baseline (Pure Metric Propagation)

Current  $\Lambda$ CDM Paradigm: Precision cosmology anticipates that the Hubble constant derived from gravitational wave standard sirens (e.g., binary neutron star mergers) will eventually converge statistically with the  $\Lambda$ CDM consensus derived from standard candles (photonic measurements such as Cepheids and Type Ia Supernovae).

Topological Framework Prediction: Standard optical measurements rely on photons, which interact electromagnetically and thermodynamically with the intergalactic baryonic medium, thereby experiencing a macroscopic "friction" that convolutes the perceived kinematic expansion. Gravitational waves, however, are pure kinematic ripples propagating exclusively through the topological manifold itself, remaining completely unaffected by photonic-baryonic resistance.

Empirical Falsification Test: As the sample size of standard sirens measured by next-generation gravitational wave observatories (LIGO/Virgo/KAGRA, LISA) increases across cosmological distances, their derived Hubble constant will systematically fail to converge with the lower  $\Lambda$ CDM consensus (67-73 km/s/Mpc). Instead, by isolating the pure topological tension of the metric, standard sirens will exhibit a statistically significant upward deviation, trending explicitly toward the pristine unhindered baseline of  $H_{base} = 81.17$  km/s/Mpc.

**Table 1: Topological Spacetime Cosmology vs. Standard  $\Lambda$ CDM Predictions**

| <b>Empirical Test / Environment</b>       | <b>Standard Model (<math>\Lambda</math>CDM) Expectation</b>                            | <b>Topological Spacetime Prediction</b>                                                         | <b>Target Observatories / Surveys</b> |
|-------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Deep Cosmic Voids</b>                  | Mirrors global average ( $\sim 70$ km/s/Mpc) with minor peculiar velocity corrections. | Bypasses average entirely; converges to unhindered metric baseline $H_{base} = 81.17$ km/s/Mpc. | DESI, Euclid                          |
| <b>Directional Expansion (Anisotropy)</b> | Isotropic expansion independent of line-of-sight baryonic density.                     | Extreme anisotropy; expansion rate acts as an exact inverse function of macroscopic density.    | Euclid, Roman Space Telescope         |
| <b>Gravitational Wave Standard Sirens</b> | Ultimately converges with photonic standard candles (67-73 km/s/Mpc).                  | Trends upward toward 81.17 km/s/Mpc, isolating pure metric tension from baryonic friction.      | LIGO, Virgo, LISA                     |

## Conclusion

The standard  $\Lambda$ CDM model relies on dark parameters that are inherently untestable by direct means. By contrast, Topological Spacetime Cosmology provides a rigid, parameter-free framework that bounds cosmic acceleration to a specific metric baseline ( $H_{base} = 81.17$  km/s/Mpc). The predictions outlined above offer clear, near-term pathways to empirically falsify the uniformity of dark energy and establish baryonic mass as the sole geometric load governing the macroscopic expansion of the universe.

## References

[1] Haimovich, T. (2026). "Topological Spacetime Cosmology: Resolving the Vacuum Catastrophe, the Hubble Tension, the Dark Energy Enigma, and Deriving the Absolute Baryonic Mass of the Universe via Entropic Metric Dilation." Independent Research / Pre-print 10.5281/zenodo.22995959.